13 A particle oscillates with simple harmonic motion along a line with a maximum speed v_0.

What is the speed of the particle when the displacement of the particle is half of its amplitude?

A $\frac{1}{4}v_0$ **B** $\frac{1}{2}v_0$ **C** $\frac{3}{4}v_0$ **D** $\frac{\sqrt{3}}{2}v_0$